

Theory of Automata and Formal Languages

Recursive definition of languages

The following **three** steps are used in recursive definition

1. Some **basic strings** or words are specified in the language.
2. **Rules** for constructing more strings or words are defined in the language.
3. No strings **except** those constructed in above, are allowed to be in the language.

Examples: Defining language of **INTEGER**

Step 1: **1** is in INTEGER.

Step 2: If x is in INTEGER then $x+1$ and $x-1$ are also in INTEGER.

Step 3: No strings **except** those constructed in above, are allowed to be in INTEGER.

$L = \{,,,,,-4, -3, -2, -1, 0, 1, 2, 3, 4, ,,,, \}$

Example: Defining language of **EVEN**

Step 1: **2** is in EVEN.

Step 2: If x is in EVEN then $x+2$ and $x-2$ are also in EVEN.

Step 3: No strings except those constructed in above, are allowed to be in EVEN.

$L = \{ 2,4,6,8,10,,,,, \}$

Example: Defining the language **PALINDROME**, defined over $\Sigma = \{a,b\}$

Step 1: **a** and **b** are in PALINDROME

Step 2: if x is palindrome, then $s(x)\text{Rev}(s)$ and xx (concatenation of two palindromes) will also be palindrome, where s belongs to Σ^*

Step 3: No strings except those constructed in above, are allowed to be in palindrome.

$L = \{a, b, aa, bb, aba, bab, ,,, \}$

Example: Defining the language $\{a^n b^n\}$, $n=1,2,3,\dots$, of strings defined over $\Sigma=\{a,b\}$

Step 1: **ab** is in $\{a^n b^n\}$

Step 2: if x is in $\{a^n b^n\}$, then axb is in $\{a^n b^n\}$

Step 3: No strings except those constructed in above, are allowed to be in $\{a^n b^n\}$

As, $n=1$, starts from 1, thus, first string will be ab , for $n=0$, the first string will be λ .

Here x means a string, for example ab , or $aabb$, x means not x .

$$L = \{ ? \}$$

Example: Defining the language L, of strings **ending in a**, defined over $\Sigma=\{a,b\}$

Step 1: **a** is in Language

Step 2: if x is in Language then sx is also in Language, where s belongs to Σ^*

Step 3: No strings except those constructed in above, are allowed to be in L
a, minimum number of strings we need,

λ or b is not possible.

Here $x = a$, $s =$ all strings over $\Sigma = \{a, b\}$, means Σ^* .

$$L = \{a, bbbbaaaa, aaa, babababababaaaaaaaaaaaaaa, bbbbbba, aabba, \dots\}$$

Example: Defining the language L, of strings **beginning and ending in same letters/alphabets**, defined over $\Sigma=\{a, b\}$

Step1: a and b are in L

Step2: asa and bsb are also in L , where s belongs to Σ^* .

Step3: No strings **except** those constructed in above, are allowed to be in L.

aa and bb are the shortest strings possible in this case.

$$L = \{ ? \}$$

Example: Defining the language L, of strings **containing aa or bb**, defined over $\Sigma=\{a, b\}$.

Step1: aa and bb are in L.

Step2: $saa s$ and $sbb s$ are also in L , where s belongs to Σ^* .

Step3: No strings except those constructed in above, are allowed to be in L .

$$L = \{ ? \}$$

Example: Defining the language L, of strings containing **exactly aa**, defined over $\Sigma = \{a,b\}$.

Step1: **aa** is in L.

Step2: $saa s$ is also in L , where s belongs to b^* .

Step3: No strings **except** those constructed in above, are allowed to be in L :

Exactly aa means, neither less nor more.

 $L = \{?$

Defining Languages through **Regular Expression (RE)**

What is a Regular Expression (RE)?

Regular Expression

- Regular expressions are used to generate **patterns** (different combinations) of **Strings**.
- Regular expressions are also known as **Relational Expressions**.
- Regular expressions consist of **Operands** and **Operators**.
 - **Operands**: Example of operands are:
 - Letters (from alphabets).
 - Variables (whose values are any patterns depending upon regular expression).
 - Lambda (empty string). λ
 - Null (empty set). ϕ
 - **Operators**: Examples of operators are:
 - Union. $(...)+(...)$ or $(...)U(...)$
 - Concatenation. $(...)(...)$ or $(...) . (...)$
 - Kleene Star. r^* , $*$ means Kleene star.

General Examples of Regular Expressions (RE):

- A regular expression may be a **null** or **empty string**, $\rightarrow r = \lambda$
- A regular expression may be a letter from the input **alphabets**, $\rightarrow r = a$
[r means regular expression]
- A regular expression may be the **Union** of two regular expression, $\rightarrow r = r_1 + r_2$
- A regular expression may be **concatenation** of two regular expressions, $\rightarrow r = r_1 r_2$
- A regular expression may be **Kleene Star** of a regular expression, $\rightarrow r = r_1^*$

- A regular expression may be a regular expression in parenthesis, $\rightarrow$
 $r=(r_1)$

Now we will discuss examples of regular expression of Union, Concatenation, and Kleen Star.

- As discussed earlier, that a^* generates $\lambda, a, aa, aaa, \dots$ and a^+ generates $a, aa, aaa, aaaa, \dots$,
 So the language,
 $L_1 = \{\lambda, a, aa, aaa, aaaa, \dots\}$ infinite.
 $L_2 = \{a, aa, aaa, aaaa, \dots\}$ can easily be expressed by a^* and a^+ respectively.
- a^* and a^+ are called the Regular Express (RE) for L_1 and L_2 languages respectively.

Example1:

Now consider another language L, consisting of all possible strings, defined over $\Sigma = \{a, b\}$. This language can also be expressed by the regular expression $(a + b)^*$.

$L = \{\lambda, a, b, aa, aaa, aaaa, aaaaa, b, bb, bbb, bbbb, bbbbbb, aa, ab, ba, bb, aaa, aab, aba, abb, \dots, \dots\}$

$(a + b) (a + b) = aa, ab, ba, bb$

$(a + b) (a + b) (a + b) = aaa, aab, aba, abb, \dots$

$\Sigma = \{a, b, c\}. \quad (a + b + c)^*$

- **Question1:**
 - Define a language L, consisting of **all possible strings**, defined over $\Sigma = \{a, b\}$, also write its regular expression.
- **Solution:**
 - If $\Sigma = \{a, b\}$, then
 - $\Sigma^* = \{a, b\} = \{\lambda, a, b, aa, ab, ba, bb, \dots\}$
 - RE = $(a + b)^*$.

- **Explanation:**

- $(a + b)^*$
- for $*$ = 0, $(a + b)^* = \{ \lambda \}$; Means, 0 length strings.
- for $*$ = 1, $(a + b)^* = (a+b) = \{a, b\}$; Means, 1 length strings.
- for $*$ = 2, $(a + b)^* = (a+b) (a+b) = \{aa, ab, ba, bb\}$; Means, 2 length strings.
- for $*$ = 3, $(a + b)^* = (a+b) (a+b) (a+b)$; Means, 3 length strings.
- for $*$ = n, $(a + b)^* = (a+b) (a+b) (a+b), \dots$, up to n ; Means, n length strings.

Example2:

Now consider another language L, of strings having **exactly aa**, defined over $\Sigma = \{a, b\}$, then it's regular expression may be **$b^*aa b^*$** .

- **Question2:** Define a language L, of strings aa, defined over $\Sigma = \{a, b\}$, write its regular expression.
- **Solution:** RE = $b^*aa b^*$ (aa b^* , b^*aa are not correct)
 - $L = \{aa, aab, aabb, aabbb, \dots, baa, bbaa, bbbbaa, \dots, baa, baab, baabb, baabbb, \dots, baab, baabb, baabbb, baabbbb, \dots\}$
- **Note:** There may be more than one RE for a language.

Example3:

Now consider another language L, of **even length**, defined over $\Sigma = \{a, b\}$, then it's regular expression may be **$((a+b)(a+b))^*$**

For $*$ = 0, we have λ

For $*$ = 1, we have string of length = 2, aa, ab, ba, bb

For $*$ = 2, we have strings of length = 4,

- **Question3:** Define a language L, of strings of **even-length**, defined over $\Sigma = \{a, b\}$, write its regular expression.
- **Solution:** $((a+b)(a+b))^*$
 - For $*$ = 0, $L = \{ \lambda \}$,
 - For $*$ = 1, RE = $(a+b) (a+b)$, $L = \{aa, ab, ba, bb\}$; Means, two lengths even strings.

- For $*$ = 2, RE = $(a+b)(a+b)(a+b)(a+b)$, $L = \{aaaa, aaab, \dots, bbbb\}$; Means, four lengths even strings.
- For $*$ = 3, RE = $(a+b)(a+b)(a+b)(a+b)(a+b)(a+b)$; Means, six lengths even strings.
- Ans so-on.

Example4: Now consider another language L, of **odd length**, defined over $\Sigma = \{a, b\}$, then it's regular expression may be

$(a+b)((a+b)(a+b))^*$ or $((a+b)(a+b))^*(a+b)$

- **Question4:** Define a language L, of strings of **odd length**, write its RE.
- **Solution:** RE = $(a+b)((a+b)(a+b))^*$ or $((a+b)(a+b))^*(a+b)$
- L = ?? (do your self).
- **Explanation:**
 - All strings having odd length, here we discussed only two possibilities of RE, there may be other possibilities for this type of language.
 - $((a+b)(a+b))^*$ regular expression will generate even length strings,
 - $(a+b)$ regular expression will generate odd length strings.
 - Concatenation of both these regular expressions will generate odd length strings.
 - We can write more than one RE for any language.
 - However, from one RE, only one language can be generated.
- **Remark:** It may be noted that, a language may be expressed by more than one regular expressions, while a given regular expression, there exist a unique language, generated by that regular expression.

Example 5: Consider the language, defined over $\Sigma=\{a, b\}$ of strings having **at least one a**, may be expressed by a regular expression $(a+b)^*a(a+b)^*$.

- **Question5:** Define a language L, of strings **having at least one a**, defined over $\Sigma=\{a, b\}$, write its RE.
- **Solution:**
 - RE = $(a+b)^*a(a+b)^*$.

- $L = \text{do yourself. } \{a, aa, ab, ba, abababa, \}$

- **Explanation:** In this case, we have three possibilities, a at start, a in middle, and a at the end of strings. $(a+b)^*a(a+b)^*$ will generate all three types of strings.

Example 6: Consider the language, defined over $\Sigma = \{a, b\}$, of strings having **at least one a and one b**, may be expressed by the following regular expression.

- **Explanation:** $(a+b)^*a(a+b)^*b(a+b)^* + (a+b)^*b(a+b)^*a(a+b)^*$

it seems very lengthy, see the + in middle, thus it is sum to two regular expressions,

see the first part, $(a+b)^*a(a+b)^*b(a+b)^*$, if first $(a+b)^*$ is equal to λ , then the strings will start with a,,,,

see the second part, $(a+b)^*b(a+b)^*a(a+b)^*$, if the first $(a+b)^*$ is equal to λ , then the strings will start with b.....

- **Question:** Define a language L, of strings, having **at least one a and one b**, defined over $\Sigma = \{a, b\}$, also write its regular expression.

- **Solution:**

$$RE = (a+b)^*a(a+b)^*b(a+b)^* + (a+b)^*b(a+b)^*a(a+b)^*$$

$$L = \{ \text{do your self} \}$$

Example7: Consider the language, defined over $\Sigma = \{a, b\}$, of strings, **starting with aa and ending with bb**.

$$aa(a+b)^*bb$$

- **Question:** Define a language L, defined over $\Sigma = \{a, b\}$, of strings, **starting with aa and ending with bb**. Also write its RE.

- **Solution:**

$$RE = aa(a+b)^*bb$$

$$L = \{ \text{do your self} \}.$$

Example8:

- **Question:** Define a language L, of strings, **start with a and ending with b or starting with b and ending with a**, defined over $\Sigma = \{a,b\}$, also write its RE.

$a(a+b)^*b + b(a+b)^*a$

$a(b+a)^*b + b(b+a)^*a$

- **Solution:**

RE = $a(a+b)^*b + b(a+b)^*a$

L = { do your self }.

Example9:

- **Question:** Define a language L, of strings, **starting with a**, defined over $\Sigma = \{a,b\}$, also write its RE.

$a(a+b)^*$

$a(b+a)^*$

- **Solution:**

RE = $a(a+b)^*$

L = { do your self }.

Example10:

- **Question:** Define a language L, of strings, **starting and ending in same letter**, defined over $\Sigma = \{a,b\}$, also write its RE.

$(a+b) + a(a+b)^*a + b(a+b)^*b$

L+ {a, b,

- **Solution:**

$$RE = (a+b)^+ a (a+b)^* a + b (a+b)^* b$$

$L = \{ \text{do your self} \}.$

Example11:

- **Question:** Define a language L , of strings, **not ending in a**, defined over $\Sigma = \{a,b\}$, also write its RE.

- **Solution:**

$$RE = (a+b)^* b + \lambda \quad \text{or} \quad ((a+b)^* b)^*$$

$L = \{ \text{do your self} \}.$

Example12:

- **Question:** Define a language **EVEN-EVEN**, of strings, defined over $\Sigma = \{a,b\}$, having even number of a's and even number of b's, also write its RE.

$L = \{ \lambda, aa, bb, \text{aa}aa, aabb, bbbb, bbaa, \text{ab}ab, baba, abba, baab, \dots \}$

- **Solution:**

$$RE = (aa + bb + (ab+ba)(aa+bb)^*(ab+ba))^*$$

- $L = \{ \text{do your self} \}.$

Example13:

- **Question:** What is the difference between the following two regular expressions:
 - $r_1 = a^* + b^*$
 - $r_2 = (a+b)^*$
- **Solution:** r_1 cannot generate any string of concatenation of a and b, while r_2 can generate such strings. When using r_1 ; strings will consist of only a's or b's or λ .

Equal Regular Expressions:

For **one language**, there may be **more than one regular expressions**, these regular expressions are called **equivalent** regular expressions.

Equivalent Regular Expression

- **Definition:** Two regular expressions are said to be equivalent, if they generate the same language.
- **Example:** consider the following two regular expressions,

$$r_1 = (a+b)^* (aa+bb)$$

$$r_2 = (a+b)^* aa + (a+b)^* bb$$

The above two regular expressions, define a language of **strings ending in aa or bb**.

Note: if $r_1 = (aa+bb)$ and $r_2 = (a+b)$, then

- $r_1 + r_2 = (aa+bb) + (a+b)$
- $r_1 r_2 = (aa+bb) (a+b) = (aaa + aab + bba + bbb)$
- $r_1^* = (aa+bb)^*$